

Eduardo D. Sontag

Northeastern University, Boston

<http://sontaglab.org>

A - Background

E. Sontag's work spans control and dynamical systems theory, theoretical computer science, machine learning, cancer and immunology, and molecular, synthetic, and computational biology. His work in nonlinear control theory led in the late 1980s to mid 1990s to the introduction of the concept of input-to-state stability (ISS), a stability theory notion for nonlinear systems, and various related variants (iISS, etc.), as well as in 1981 the first formulation of control-Lyapunov functions, and later a “universal formula” for smooth stabilization. At the time, he also developed tools for stabilization of linear systems under actuator saturation. In the 1980s, he pioneered techniques for the analysis of nonlinear systems based on commutative algebra and algebraic geometry, for observability and minimal realizations. In the early to mid 1980s, he developed notions of computational complexity for control systems, and published one of the first papers on hybrid (piecewise linear) systems. In the late 1990s, Sontag turned to a focus on discontinuous stabilization, a topic that had originated in joint work with Sussmann in 1980. Other work in the 1980s to early 1990s dealt with the foundations of observation spaces, identifiability, and input/output equations for nonlinear behaviors, and the finalizing of a major textbook in Mathematical Control Theory.

Sontag has a long-standing interest in biology and neuroscience, starting with a book on AI that he authored while an undergraduate. In the 1990s, he established basic results on representability, identifiability, and super-Turing computability by neural networks, as well as “PAC learning” sample estimates for dynamical systems, rates of approximation in function approximation, and a foundational result on neural-network feedback control (“two hidden layers are necessary”). Since the year 2000, he has increasingly been motivated by molecular biology, and has published in systems biology (and more pure biology) journals on the topics of bacterial chemotaxis, cell cycle, immune recognition, interactions between cancers and infections, chemotherapy-induced resistance, metastasis, epigenetics, ribosome flow models, and tumor heterogeneity. He has also worked on population models (Lotka-Volterra species competition, epidemiology). Another recent major direction of work has been in synthetic biology, and specifically the design of genetic and post-translational circuits for gene copy number compensation, rejection of disturbances, and Boolean computation. More generally in systems biology, he has worked on fundamental mathematical principles of monotone systems, stochastic models, and chemical network theory.

B - Professional Preparation

- U. of Buenos Aires, Buenos Aires, Mathematics, Licenciado, 1972
- U. of Florida, Gainesville, Mathematics, Ph.D., 1976 (advisor: R.E. Kalman)

C - Appointments

- 2018- *Northeastern*: Distinguished University Professor, Northeastern University, Electrical and Computer Engineering and Bioengineering (2018-); Affiliate Professor, Mathematics (2019-) and Chemical Engineering (2019-); Faculty and Leadership Council, Institute for Experiential AI (2021-)
- 2006- *MIT*: Research affiliate.

- 2018- *Harvard Medical School*: Faculty, Laboratory for Systems Pharmacology.
- 1977-2017 *Rutgers*: Distinguished Professor of Mathematics (1990-2017); Professor (1987-1990); Associate Professor (1982-1987); Assistant Professor (1977-1982); Member of Graduate Faculty, Computer Science (1985-2017); Director, Center for Systems and Control (1988-2017); Member of Graduate Faculty, Electrical & Computer Engineering (1989-2017); Director, Biomathematics Undergraduate Interdisciplinary Degree (2002-2017); Founding Member & Advisory Committee, BioMaPS Institute for Quantitative Biology (2002-2015); Member. Cancer Institute of NJ (2008-2017); Director, Center for Quantitative Biology (2015-2017); Graduate Director, Institute for Quantitative Biomedicine (2015-2017)

D - Selected Honors and Awards

- *Major Research Awards*:
 - ◊ Invited Address to International Congress of Mathematicians (IMU, 1994)
 - ◊ Reid Prize in Applied Mathematics (SIAM, 2001)
 - ◊ Board of Trustees Award for Excellence in Research (Rutgers, 2002)
 - ◊ Bode Prize (IEEE, 2002)
 - ◊ Field Award in Control Systems (IEEE, 2011).
 - ◊ Bellman Control Heritage Award (AIAA/AICHe/Informs/ASCE/ASME/IEEE/ISA/SIAM2022)
- *Professional Society Fellow Awards*: Institute of Electrical and Electronics Engineers (IEEE) (1993), Society for Industrial and Applied Mathematics (SIAM) (2009), American Mathematical Society (AMS) (2013), International Fed. of Automatic Control (IFAC) (2014).
- *Teaching and Mentorship Awards*: Teacher/Scholar Award (Rutgers, 2005).
- *Selected Public Research Rankings*:
 - ◊ Research.com: top 100 US electrical engineers and top 100 US mathematicians
<https://research.com/scientists-rankings/electronics-and-electrical-engineering/us?page=1>
<https://research.com/news-events/world-ranking-of-top-mathematics-scientists-2022>
 - ◊ Most-cited author in: IEEE Transactions on Automatic Control in 1981, 1996, 1997; Systems and Control Letters 1989, 1991, 1995, 1998, and lifetime of journal; SIAM Journal on Control and Optimization 1983, 1986; Theoretical Computer Science 1994; as well as many other journal/years
<https://exaly.com/author/1446921/eduardo-d-sontag/ranking>
 - ◊ Elsevier/Stanford: top 0.5% among 2% top scientists
<https://elsevier.digitalcommonsdata.com/datasets/btchxktzyw/3>
 - ◊ MathScinet: most-cited applied mathematician who got PhD in 1976
<https://mathcitations.github.io/>

E - Publications

- h-index 104, 60K citations, mostly single or two-authored (Google Scholar June 2022)
<https://scholar.google.com/citations?user=CCToMygAAAAJ&hl=en>
 URL for all reprints: <http://www.sontaglab.org/publications.html>